

W1

Algebra For Honor Math

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1 Assignment A1

1.1 Problem 1

(I) **Prove that $\sqrt{3}$ is irrational.**

We need to prove a lemma first before we prove the proposition.

Lemma 1. *For all $n \in \mathbb{Z}$, if $3 \mid n^2$, then $3 \mid n$.*

Proof. Suppose $3 \nmid n$. There are two cases to discuss.

case 1: $n = 3k + 1$

According to the definition of divisibility, we have

$$n = 3k + 1 \quad \exists k \in \mathbb{Z}.$$

Taking the square of both sides, it follows that

$$n^2 = (3k + 1)^2 = 9k^2 + 6k + 1 = 3c + 1 \quad c = 3k^2 + 2k \in \mathbb{Z}.$$

thus, $3 \nmid n^2$.

case 2: $n = 3k + 2$

According to the definition of divisibility, we have

$$\exists k \in \mathbb{Z} \text{ s.t. } n = 3k + 2$$

Taking the square of both sides, it follows that

$$n^2 = (3k + 2)^2 = 9k^2 + 12k + 4 = 3c + 1 \quad c = 3k^2 + 4k + 1 \in \mathbb{Z}.$$

thus, $3 \nmid n^2$. ■

Here is the proof of the question:

Proof. Suppose $\sqrt{3} \in \mathbb{Q}$.

Then, according to the definition of rational numbers,

$$\exists m, n \in \mathbb{Z}, n \neq 0 \quad \text{s.t.} \quad \sqrt{3} = \frac{m}{n}.$$

We assume that m and n have no common factor other than 1. Taking the square of the both sides, we get

$$3 = \frac{m^2}{n^2} \quad \implies \quad m^2 = 3n^2,$$

which implies that $3 \mid m$ according to the lemma 1. According to the definition of divisibility,

$$\exists k \in \mathbb{Z} \quad \text{s.t.} \quad m = 3k.$$

Substituting $m = 3k$ back into $m^2 = 3n^2$, we have

$$\begin{aligned} n^2 &= (3k)^2/3 \\ &= 9k^2/3 \\ &= 3d, d = k^2 \in \mathbb{Z}. \end{aligned}$$

It means that $3 \mid n$.

Since $3 \mid n$ and $3 \mid m$, it contradicts the assumption that n and m have no common factor other than 1.

Therefore, $\sqrt{3} \in \mathbb{R} \setminus \mathbb{Q}$. ■

(II) *Proof.* We define

$$P := \{x \in \mathbb{R} : x > 0\}$$

to be a nonempty subset of $\mathbb{R}$ that satisfies Trichotomy law, Closure under addition, and closure under multiplication.

Suppose $a, b \in P$. There are two cases to discuss, which are

- $a > b$.
- $a = b$.
- $b < a$.

We will prove case $a > b$ first.

According to the given statement, we have

$$(\forall c > 0) (\exists x > 0) x^2 = c.$$

It follows that there $\exists a', b' \in P$ such that

$$(a')^2 = a \quad \text{and} \quad (b')^2 = b.$$

Since $a > b$, it means that

$$\begin{aligned} a - b &= (a')^2 - (b')^2 \\ &= (a' - b')(a' + b') \in P. \end{aligned}$$

We know that $a' + b' \in P$, so $a' - b' \in P$.

By the ordered field properties, we have

$$(a' - b')^2 \in P \implies (a' - b')^2 > 0.$$

given $a' - b' \in P$ and closure under multiplication.

Expanding the left side, we get

$$(a')^2 - 2a'b' + (b')^2 > 0 \implies \frac{(a')^2 + (b')^2}{2} > a'b'. \quad (1.1)$$

By substituting $(a')^2 = a$ and $(b')^2 = b$ into equation (1.1), we have

$$\frac{a + b}{2} > a'b'.$$

Since $a' \in P$ and $b' \in P$, it follows from the closure under multiplication that

$$a'b' \in P \implies a'b' > 0.$$

Moreover,

$$(a'b')^2 = (a')^2 (b')^2 = ab \in P \implies ab > 0$$

Since $a'b' > 0$ and $ab = a'b'$, therefore, ab is a positive square root of $a'b'$.

$$a'b' = \sqrt{ab}.$$

is the only option.

As a result,

$$\frac{a+b}{2} > \sqrt{ab}.$$

For $b > a$, it can be proved using similar way as $a > b$ by swapping a with b and b with a .

For $a = b$, we have

$$\frac{a+b}{2} = \frac{2a}{2} = a = \sqrt{a^2} = \sqrt{ab}.$$

Therefore, we get

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

Thus, the inequality has been proved. ■

Reference: The ordered field properties were consulted in V. A. Zorich, *Mathematical Analysis I*, Section 2.2. Prove style is modified based on the suggestions from my dear father, Changwei Hu.

1.2 Problem 2

1.2.1 (I)

Definition 1. A set $S \subseteq \mathbb{R}$ is called inductive if $1 \in S$ and $k \in S$ implies $k + 1 \in S$.

$\mathbb{R}$ is inductive since $1 \in \mathbb{R}$ and for all $x \in \mathbb{R}$, $x + 1 \in \mathbb{R}$.

$(0, \infty)$ is inductive since $1 \in (0, \infty)$ and for all $x \in (0, \infty)$, $x + 1 \in (0, \infty)$.

$(0, \infty) \setminus \{5\}$ is not inductive since $4 \in (0, \infty) \setminus \{5\}$ but $4 + 1 = 5 \notin (0, \infty) \setminus \{5\}$.

1.2.2 (II)

Proof. Suppose that S and T are inductive sets, and $U = S \cap T$.

By the definition of inductive set, we have

$$1 \in S \quad \text{and} \quad k \in S \implies k + 1 \in S. \quad (1.2)$$

and

$$1 \in T \quad \text{and} \quad k \in T \implies k + 1 \in T. \quad (1.3)$$

Since 1 is in both S and T , it means that $1 \in U$.

By statements (1.2) and (1.3), we get

$$k \in U \iff (k \in S \wedge k \in T) \implies (k + 1 \in S \wedge k + 1 \in T) \iff k + 1 \in U.$$

Thus, U is an inductive set. ■

1.2.3 (III)

Proof. Let $\mathcal{I}$ be the collection of all inductive subsets. We define

$$\mathbb{N} = \bigcap_{S \in \mathcal{I}} S.$$

Since

$$(\forall S \in \mathcal{I}) 1 \in S,$$

it follows that $1 \in \mathbb{N}$.

It suffices to show that

$$k \in \mathbb{N} \implies k + 1 \in \mathbb{N}.$$

Suppose $k \in \mathbb{N}$. By the definition of the intersection of sets, it means that

$$(\forall S \in \mathcal{I}) k \in S.$$

It follows the definition of inductive set that

$$(\forall S \in \mathcal{I}) k + 1 \in S,$$

which implies

$$k + 1 \in \mathbb{N}.$$

Thus, $\mathbb{N}$ is inductive. ■

1.3 Problem 3

1.3.1 (I)

Proof. Suppose that $P(1)$ is true and

$$(\forall n \in \mathbb{N}) [(P(1) \wedge P(2) \wedge \cdots \wedge P(n)) \implies P(n+1)].$$

Let

$$T(n) := P(1) \wedge P(2) \wedge \cdots \wedge P(n).$$

Since $P(1)$ is true, it implies that $T(1)$ is also true. We also assume $T(n)$ is true.

By the assumption of strong induction, we obtain

$$T(n) \implies P(n+1),$$

and hence, since $T(n)$ is already assumed to be true,

$$T(n) \wedge P(n+1).$$

By the definition of $T(n+1)$, it follows that

$$T(n) \implies T(n+1).$$

Following the Principle of Mathematical Induction, this yields

$$(\forall n \in \mathbb{N}) T(n) \implies (\forall n \in \mathbb{N}) P(n).$$

Thus, Principle of Strong Mathematical Induction is correct. ■

1.3.2 (II)

Proof. We define S is a set such that

$$S := \{n \in \mathbb{N} \cup \{0\} : p_n(x) \text{ is a degree } n \text{ polynomial and the coefficient of } x^n \text{ in } p_n(x) \text{ is } 1\}$$

We know $0 \in S$ is true since $p_0(x) = 1x^0 = 1 \cdot 1 = 1$ is a degree 0 polynomial and the coefficient of x^0 is 1.

We also know that $1 \in S$ is true since $p_1(x) = 1 \cdot x$ is a degree 1 polynomial and the coefficient of x is 1.

It is suffice to show that

$$(n-1 \in S) \wedge (n \in S) \implies (n+1 \in S).$$

Suppose $n-1 \in S$ and $n \in S$ are true. We define

$$p_{n-1}(x) = x^{n-1} + q(x) \quad \text{and} \quad p_n = x^n + r(x),$$

where

$$\deg r \leq n-1 \quad \text{and} \quad \deg q \leq n-2.$$

Then, by the given statement

$$\begin{aligned} p_{n+1}(x) &= x \cdot p_n(x) - p_{n-1}(x) \\ &= x \cdot (x^n + r(x)) - (x^{n-1} + q(x)) \\ &= x^{n+1} + xr(x) - x^{n-1} - q(x). \end{aligned}$$

Since the term $xr(x) - x^{n-1} - q(x)$ has at most degree n , the term x^{n+1} cannot be canceled. Therefore, $p_{n+1}(x)$ is a degree $n+1$ polynomial and the coefficient of x^{n+1} is 1

Thus, by the Principle of Complete Mathematical Induction,

$$\forall n \in \mathbb{N} \cup \{0\}, n \in S,$$

which implies that the given statement is true. ■

1.3.3 III

Proof. Let $0 < |t| < 1$ and define statement

$$T(n) : p_n(t + t^{-1}) = \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}}.$$

We know that $T(0)$ is true since

$$p_0(t + t^{-1}) = \frac{t^{0+1} - t^{0-1}}{t - t^{-1}} = \frac{t - t^{-1}}{t - t^{-1}} = 1.$$

We also know that $T(1)$ is true since

$$p_1(t + t^{-1}) = \frac{t^{1+1} - t^{-1-1}}{t - t^{-1}} = \frac{t^2 - t^{-2}}{t - t^{-1}} = \frac{(t^1 - t^{-1})(t^1 + t^{-1})}{t - t^{-1}} = t + t^{-1}.$$

Suppose that $T(n-1)$ and $T(n)$ are true. It suffices to show that

$$(T(n-1) \wedge T(n)) \implies T(n+1).$$

Following from the given statement,

$$\begin{aligned} p_{n+1}(t + t^{-1}) &= (t + t^{-1}) p_n(t + t^{-1}) - p_{n-1}(t + t^{-1}) \\ &= (t + t^{-1}) \cdot \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}} - \frac{t^{n-1+1} - t^{-(n-1)-1}}{t - t^{-1}} \\ &= (t + t^{-1}) \cdot \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}} - \frac{t^n - t^{-n}}{t - t^{-1}} \\ &= \frac{t^{n+2} - t^{-n+2}}{t - t^{-1}} \\ &= \frac{t^{(n+1)+1} - t^{-(n+1)-1}}{t - t^{-1}} \end{aligned}$$

This implies $T(n+1)$ is true.

By the Principle of Complete Mathematical Induction, we get

$$(\forall n \in \mathbb{N} \cup \{0\})T(n).$$

Thus,

$$(\forall n \in \mathbb{N} \cup \{0\}) \left[p_n(t + t^{-1}) = \frac{t^{n+1} - t^{-n-1}}{t - t^{-1}} \right].$$

■

1.3.4 IV

Proof. Given function

$$g : D \rightarrow \mathbb{R}, \quad g(t) := t + t^{-1}$$

where

$$D := \{x \in \mathbb{R} : 0 < |x| < 1\}.$$

Let $y \in \text{rng}(g)$, then

$$(\exists x \in D) y = t + t^{-1}.$$

Since $t \neq 0$, we get

$$y = x + \frac{1}{x} \iff xy = x^2 + 1 \iff 0 = x^2 - yx + 1.$$

By the definition of the set D , element d must be in real number set. So the equation

$$0 = x^2 - yx + 1$$

must have real roots. Therefore,

$$(-y)^2 - 4 \cdot (1) \cdot (1) \geq 0 \implies y^2 - 4 \geq 0 \implies y^2 \geq 4 \implies |y| \geq 2.$$

However, $|y| \neq 2$ since

$$t = \frac{-(-y) \pm \sqrt{(-y)^2 - 4 \cdot a \cdot 1}}{2 \cdot 1} = \frac{-(2) \pm \sqrt{(-2)^2 - 4 \cdot a \cdot 1}}{2 \cdot 1} = \frac{-2}{2} = -1$$

and

$$t = \frac{-(-y) \pm \sqrt{(-y)^2 - 4 \cdot a \cdot 1}}{2 \cdot 1} = \frac{-(-2) \pm \sqrt{(2)^2 - 4 \cdot a \cdot 1}}{2 \cdot 1} = \frac{2}{2} = 1$$

which makes $x \notin D$. Thus, $|y| > 2$, which also implies $\text{rng}(g) = (-\infty, -2) \cup (2, \infty)$.

We want to show that

$$(\forall t \in (-1, 0) \cup (0, 1)) p_n(t + t^{-1}) \neq 0$$

.

Suppose, for contradiction, $p_n(t + t^{-1}) = 0$. We have

$$\begin{aligned}\frac{t^{n+1} - t^{-n-1}}{t - t^{-1}} &= 0 \\ \Rightarrow t^{n+1} - t^{-n-1} &= 0 \\ \Rightarrow t^{n+1} &= t^{-n-1} \\ \Rightarrow t^{2n+2} &= t^0 = 1\end{aligned}$$

Since $2n + 2 > 0$ and $0 < |t| < 1$,

$$0 < |t|^{2n+2} < 1.$$

This is a contradiction since t will never be 1.

Thus, for all $t \in (-1, 0) \cup (0, 1)$, $p_n(t + t^{-1}) \neq 0$.

Since $x = g(t) = t + t^{-1}$, this yields

$$(\forall x \in \text{rng}(g))p_n(x) \neq 0 \quad \implies \quad ((\forall x \in \mathbb{R} \setminus [2, 2])p_n(x) \neq 0)$$

It follows that

$$(\forall x \in [2, 2])p_n(x) = 0$$

Thus, the claim has been proved. ■

1.4 Problem 4

1.4.1 I

Proof. Suppose

$$(\forall x > 0) (\exists n \in \mathbb{N}) \frac{1}{n} < x.$$

Multiplying both sides by $\frac{n}{x}$,

$$\frac{1}{x} < n.$$

Since every positive real number has a positive multiplicative inverse we have

$$(\forall x > 0) (\exists x' > 0) x \cdot x' = 1. \implies x' = \frac{1}{x} > 0$$

This yields

$$(\forall x > 0) (\exists x' > 0) (\exists n \in \mathbb{N}) \frac{1}{x} = x' < n,$$

Now, Since we also have

$$(\forall x' > 0) (\exists x > 0) x' = \frac{1}{x} > 0,$$

this yields

$$(\forall x' > 0) (\exists n \in \mathbb{N}) x' < n.$$

Thus, the Archimedean property is proved. ■

1.4.2 ii

Proof for the case for $x \in \mathbb{Q}$

We need to prove

$$(\forall x \in \mathbb{Q}) \left[(x > 0) \implies (\exists n \in \mathbb{N}) \frac{1}{n} < x \right].$$

Proof. Since $x \in \mathbb{Q}$, by the definition

$$(\forall x \in \mathbb{Q}) (\exists m, d \in \mathbb{Z}) \left[d \neq 0 \implies x = \frac{m}{d} \right].$$

Since $x > 0$, which implies $m \neq 0$.

Consider the case where both m and d are positive first. We have

$$\frac{1}{n} < \frac{m}{d} \implies d < mn.$$

Since $d \geq 1$ and $m \geq 1$, we can set

$$n = d + 1.$$

This yields,

$$d \leq md < m(d + 1) = md + m.$$

Therefore,

$$d < mn \implies \frac{1}{n} < \frac{m}{d}.$$

Thus, the original claim holds.

Consider the case where $m, d < 0$.

Since

$$\frac{m}{d} = \frac{-m}{-d},$$

we are able to make

$$m' = -m \quad \text{and} \quad d' = -d$$

Then treat them as the case where m and d are both positive. ■

Prove for the case where $x > 0, x = \sqrt{y}, \exists y \in \mathbb{Q}$. We need to prove

$$(\forall x > 0) \left[[(\exists y \in \mathbb{Q}) [y > 0 \wedge x = \sqrt{y}]] \implies (\exists n \in \mathbb{N}) \frac{1}{n} < x \right].$$

Proof. Let

$$y = \frac{m}{d}, \exists m \in \mathbb{Z}, \exists d \in \mathbb{Z}, d \neq 0.$$

Since $y > 0$, which implies

$$m \neq 0,$$

and we have two cases to discuss. They are

$$(m \geq 1) \wedge (d \geq 1) \quad \text{and} \quad (m \leq -1) \wedge (d \leq -1).$$

- Case $(m \geq 1) \wedge (d \geq 1)$:

$$\begin{aligned} (\exists n \in \mathbb{N}) \frac{1}{n} < x &\Leftrightarrow (\exists n \in \mathbb{N}) \frac{1}{n} < \sqrt{y} = \sqrt{\frac{m}{d}} \\ &\Leftrightarrow (n \in \mathbb{N}) \frac{1}{n^2} < \frac{m}{d} \\ &\Leftrightarrow (\exists n \in \mathbb{N}) d < mn^2, \end{aligned}$$

Since $m \geq 1$, we can select $n = (d + 1)$. We have

$$mn^2 = m(d + 1)^2 = m(d^2 + 2d + 1) = md^2 + 2md + m > d$$

Thus, the claim hold.

- Case $(m \leq -1) \wedge (d \leq -1)$

Since $y > 0$, we are able to multiple y by $\frac{-1}{-1} = 1$, where we have

$$y = \frac{m}{d} \cdot 1 = \frac{m}{d} \cdot \frac{-1}{-1} = \frac{-m}{-d}, -m \geq 1, -d \geq 1.$$

Then, we treat this case similar to the last case.

Thus, the claim holds. ■

1.4.3 III

Proof. Suppose x is an arbitrary positive number in $\mathbb{R}$. We define the set

$$T := \{m \in \mathbb{N} : x < m\}.$$

By Archimedean Property, we know that $T \neq \emptyset$. By well ordering principle, since $T \neq \emptyset$ and $T \subset \mathbb{N}$, T has a least element. We define

$$k := \min T.$$

Since k is the least element in T , which means

$$k > x \quad \text{and} \quad k - 1 \notin T.$$

The definition of set T implies

$$k - 1 \leq x.$$

Thus, we have

$$k - 1 \leq x < k.$$

Let $n = k - 1$, we replace k by n .

$$n \leq x < n + 1.$$

■